Data-Driven and Physics-Informed Neural Networks

1. Data, protocol, and model selection

We compare a feedforward regression network with a jointly trained physics-informed model. Database.xls contains 1,865 samples, nine predictors, and target y. IDs are excluded from prediction. The two-row header was handled explicitly; there are no missing/nonfinite values or duplicate IDs. All observations were retained. Six pairs share all nine predictors: two also share y, while four have different y. Exact predictor combinations define groups, giving 1,859 groups and preventing repeated inputs from crossing partitions.

Seed-42 GroupShuffleSplit holds out 20% of groups: 1,493 training and 372 testing samples (80.05/19.95%). Shuffled GroupKFold, seed 42, forms five training folds with 298, 298, 300, 300, and 297 validation samples. Both models use identical saved assignments. Standardization is fitted separately on each fitting fold. Positive x1, x2, x4, x5, and x6 are optionally log10-transformed first; zero-valued features are not logged. The fixed log10(y) target accommodates its approximately 0.000075–46.863905 range. Predictions are 10 raised to the network output; invalid values raise errors, with no clipping.

The physics model predicts $m = \text{softplus}(g(X)) > 0$ jointly with $z = \log_{10}(\hat{y})$. In original numerical units, $z_{phys} = 2\log_{10}(x2) - \log_{10}(72.2) + m\log_{10}(x1)$, equivalent to $y_{phys} = x2^2/[72.2\, x1^{(-m)}]$. Its objective is $\text{mean}[(z - \log_{10}(y))^2] + \lambda\, \text{mean}[(z - z_{phys})^2] + \alpha\Sigma||W||^2$. The data-driven model omits the physics term. Biases are excluded from L2. The physical residual sends gradients to both networks; equation equivalence and nonzero gradients were verified.

Choice	Search / rationale
Prediction architecture	1–3 layers; uniform widths 16/32/64; ReLU/tanh: modest nonlinear capacities
Optimization	Adam; rates 0.0003/0.001/0.003; batches 32/64/128; L2 0/1e−5/1e−4/1e−3
Input representation	Selected logs on/off, then standardization: assess compression, especially x2/x4
Physics exponent network	Tanh hidden sizes (8), (16), (16,8); inputs x1–x2/all: limited scalar-map capacity
Physics optimization	Fixed selected backbone, batch/L2/logs; retune the three rates; λ = 0.01/0.1/1/10

Each search samples 20 distinct candidates without replacement (seed 42): 20/1,296 data-driven and 20/72 physics configurations, totaling 200 fold fits. Adam uses β=(0.9,0.999), ε=1e−7. He/ReLU and Xavier/tanh initialization are used with zero biases; output weights use Xavier initialization. Fold seeds are 42+fold. Early stopping uses validation RMSLE, a 1,000-epoch ceiling, patience 50, and min_delta=1e−5; retain the absolute minimum checkpoint. Select by mean five-fold RMSLE and report sample SD (ddof=1).

Selected data-driven settings: 16–16–16 ReLU, learning rate 0.003, batch 64, L2 0.001, logs enabled; validation RMSLE 0.299685 ± 0.016474. Physics retains that backbone and selects exponent hidden sizes [16], inputs x1_x2, learning rate 0.003, and λ=0.01; validation RMSLE 0.298574 ± 0.015089. Other features x3–x9 contribute through the prediction network.

2. Final refitting and predictive results

Before testing, durations were fixed to median selected CV best epochs: 78 (data-driven) and 116 (physics-informed). New scalers and models were fitted on all training samples, seed 42, without test/validation monitoring. $R^2 = 1 - SSE/SST$ uses original-scale residuals. $RMSLE_{10} = \sqrt{\text{mean}[(\log_{10}(1+\hat{y}) - \log_{10}(1+y))^2]}$; this +1 convention differs from the log-target training loss. Finalized models were evaluated without further adjustment.

Model	Training R^2	Training RMSLE	Testing R^2	Testing RMSLE
Data-driven	0.0896	0.2808	0.1803	0.2855
Physics-informed	0.1886	0.2725	0.2082	0.2810

3. Training behavior and learned exponent

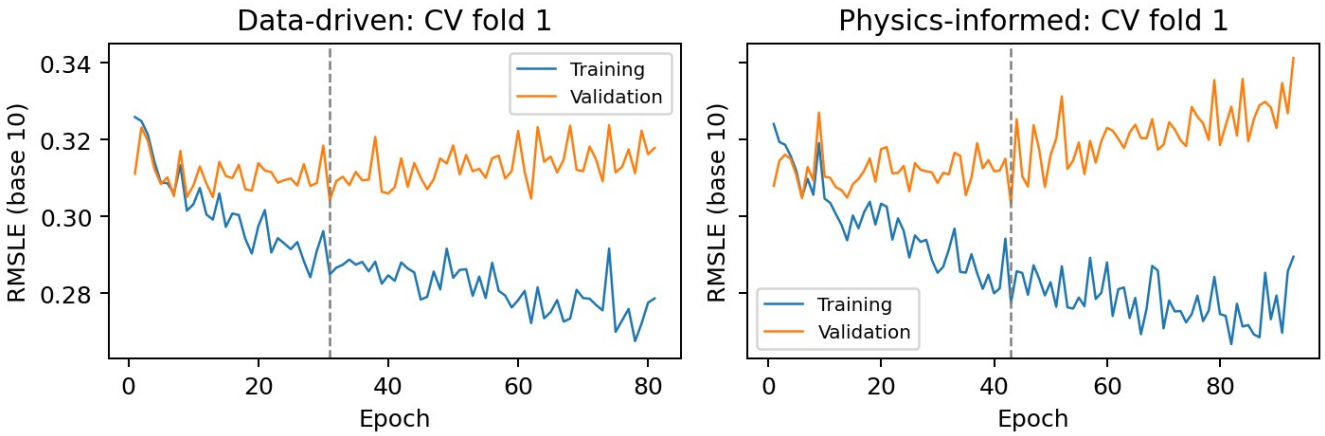


Figure 1. Training and validation RMSLE versus epoch for preselected CV fold 1 of each selected configuration. Dashed lines mark retained epochs; curves end at actual early stopping. All five folds are provided in supporting outputs. No test error was monitored against epoch.

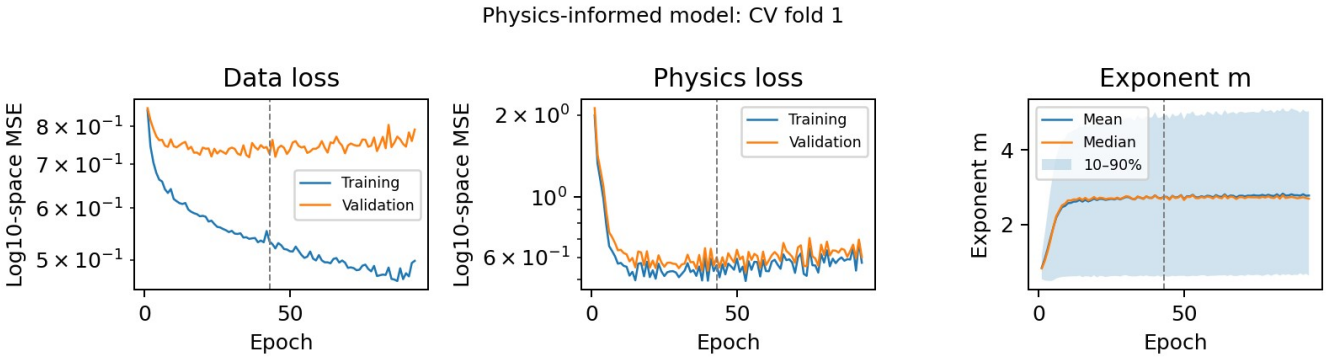


Figure 2. Separate unweighted data and physics losses (log10-space MSE), and m evolution on fixed fitting-fold inputs, for physics-informed CV fold 1. The band is the sample 10th–90th percentile, not uncertainty. Dashed lines mark the retained checkpoint. Full-training refit diagnostics are supplied separately.

m is sample-dependent; a single global exponent would be misleading. Final-model summaries below describe the learned distribution, and every sample's m is saved with its ID. Test exponents are generated from features alone. No observed y is passed to the exponent network at inference.

Partition	Mean m	Median m	SD	10th–90th percentile
Training	2.843	2.718	1.723	0.649–5.296
Testing	2.959	2.958	1.652	0.768–5.419

The physics residual RMSE is 0.7838 log10 units on training data and 0.7075 on testing data. These measure agreement of the prediction with the learned equation, not agreement of the equation with an independently established exponent. The exponent mapping and prediction network can co-adapt, so predictive accuracy and small residuals alone do not establish identifiable or causal physical parameters.

Computational record: cumulative fold-training time was 293.20 s (data-driven) and 546.87 s (physics-informed), approximately 14.00 min combined, on CPU with one PyTorch thread and deterministic algorithms. Final refits took 1.29 s and 2.67 s respectively. Timings exclude installation and smoke tests and may differ in Colab. 1 physics folds hit the epoch ceiling; 0 belonged to the selected candidate. No data-driven fit hit the ceiling. Search records include elapsed time and software versions.

4. Prediction diagnostics and interpretation

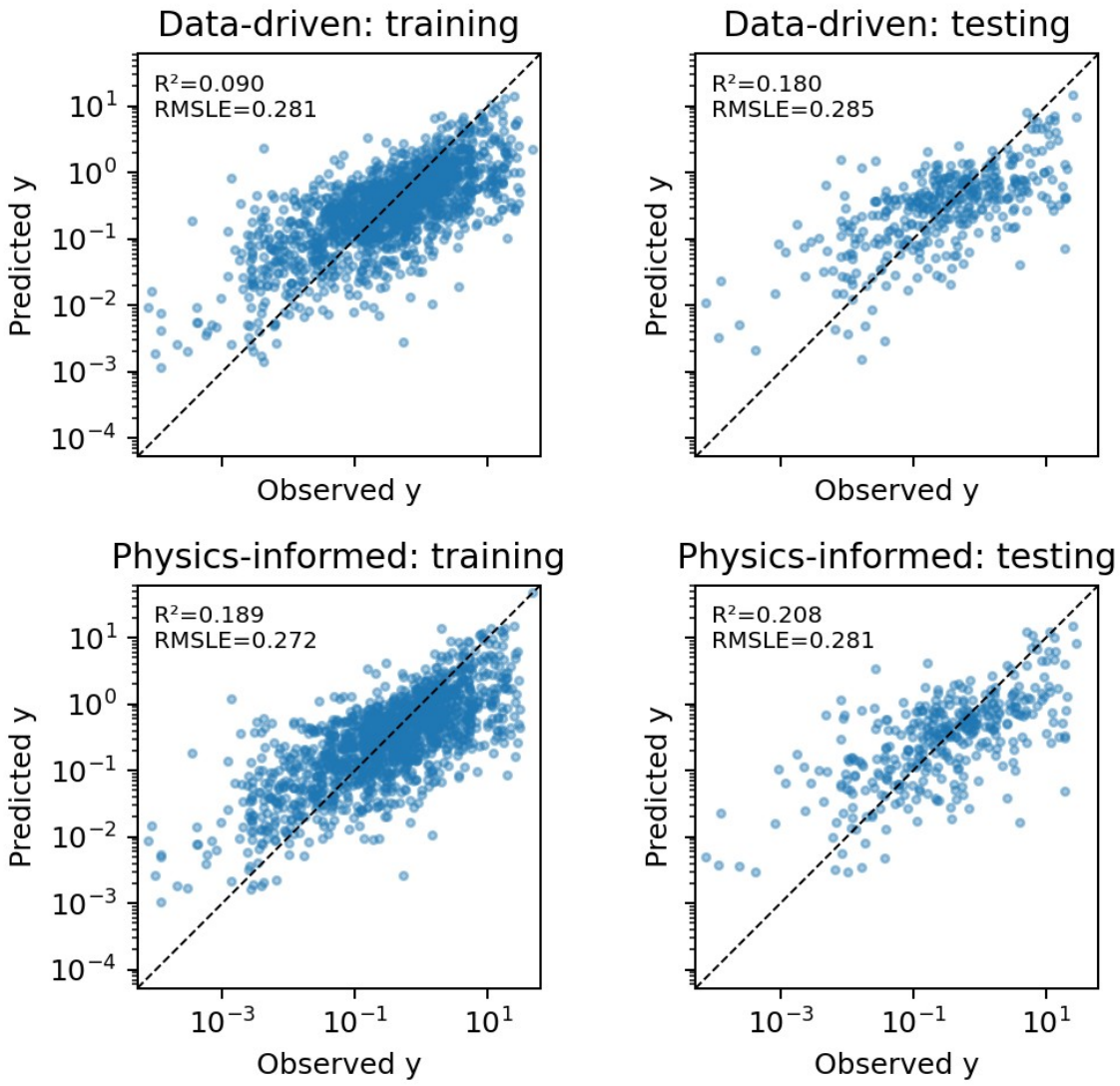


Figure 3. Final-model predictions versus observations for training and testing, with common logarithmic limits and 1:1 lines. Training points are in-sample; test points are held out. Metrics are calculated on the original predictions without clipping.

The physics-informed model has lower test RMSLE than the data-driven model (0.2810 versus 0.2855); original-scale R^2 must be considered alongside the logarithmic metric because they emphasize different errors. Both test R^2 values remain low, and the scatter shows overprediction at the smallest targets and underprediction of many large targets. The improvement on one split should not be generalized beyond this dataset. Neither training fit nor selected validation scores alone establish generalization. The saved validation tradeoff figure compares predictive RMSLE and physical residual across evaluated candidates; architectures and learning rates also vary, so it is exploratory rather than an isolated causal effect of λ.

Physical compatibility is limited: all observed x1 values are below one, so m>0 implies $y_{phys} < x2^2/72.2$. A training-only check found 359/1493 observations above this bound; they were retained. A soft physics penalty can therefore conflict with exact data fitting, and the chosen λ reflects a validation compromise rather than proof of the law. Twenty trials per model provide limited search coverage. One seed per fold does not quantify initialization uncertainty; fold SD is not a confidence interval. The physics architecture is conditioned on the selected data-driven backbone, and the fixed log-target objective differs from RMSLE's log(1+y) weighting.

Reproducibility, sources, and assistance

The notebook and Python sources reproduce preprocessing, saved partitions, both searches, refitting, evaluation, and figures. Pinned dependencies, candidate tables, fold histories/checkpoints, final settings and model checkpoints, positive predictions linked to IDs, and m summaries are supplied. See README.md; upload the provided Database.xls and run the notebook in order. Source materials are the supplied Coding Challenge.pdf and workbook; implementation uses NumPy, pandas, scikit-learn, PyTorch, and Matplotlib. OpenAI Codex assisted with implementation, execution, interpretation, figures, and report drafting; the author must review, understand, verify, and disclose this assistance.